Game Theory for Data Science

Problem Set Exam • Solutions and Grading Guide

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Grading conventions. This is a take-home exam, so we expect complete derivations. Numerical answers obtained by code without a derivation receive at most half credit for that part. Consistent carry-through of an earlier error costs at most 1 point per part.

Problem 1 (20 points)

(a) Pure equilibria: (D, C) with payoffs $(8, 3)$ and (C, D) with payoffs $(3, 8)$; (D, D) and (C, C) are not equilibria ($3 > 0$ and $8 > 6$).

Mixed: if the opponent dares with probability p, then D pays $8(1-p)$ and C pays $3p + 6(1-p) = 6 - 3p$. Indifference gives $8 - 8p = 6 - 3p$, so $p = 2/5$. Each drone dares with probability $2/5$, and each expected payoff is $8 \cdot \frac{3}{5} = \frac{24}{5} = 4.8$.

Rubric (5 pts).

- 2 pts: both pure equilibria with payoffs.
- 2 pts: mixed probability $2/5$. 1 pt: payoff $24/5$.

(b) Obedience constraints (the recommended action must be a best response to the conditional distribution of the other drone's recommendation):

$$\begin{aligned}
\text{Row told } D: &\quad 0 \cdot z_{DD} + 8 z_{DC} \;\geq\; 3 z_{DD} + 6 z_{DC} &&\iff\quad 2 z_{DC} - 3 z_{DD} \geq 0,\\
\text{Row told } C: &\quad 3 z_{CD} + 6 z_{CC} \;\geq\; 0 \cdot z_{CD} + 8 z_{CC} &&\iff\quad 3 z_{CD} - 2 z_{CC} \geq 0,\\
\text{Column told } D: &\quad 0 \cdot z_{DD} + 8 z_{CD} \;\geq\; 3 z_{DD} + 6 z_{CD} &&\iff\quad 2 z_{CD} - 3 z_{DD} \geq 0,\\
\text{Column told } C: &\quad 3 z_{DC} + 6 z_{CC} \;\geq\; 0 \cdot z_{DC} + 8 z_{CC} &&\iff\quad 3 z_{DC} - 2 z_{CC} \geq 0,
\end{aligned}$$

together with $z \geq 0$ and $z_{DD} + z_{DC} + z_{CD} + z_{CC} = 1$. (Conditioning divides both sides by the same positive probability, so it can be omitted.)

Rubric (5 pts).

- 1 pt per correct constraint (4 pts); 1 pt for the simplex constraints.
- Students who write the constraints with conditional probabilities and do not simplify: 4 of 5.

(c) Total payoff is $0 \cdot z_{DD} + 11 z_{DC} + 11 z_{CD} + 12 z_{CC} = 11 - 11 z_{DD} + z_{CC}$. Putting weight on (D, D) lowers the objective and only tightens the constraints, so $z_{DD} = 0$ at the optimum, and we maximize z_{CC} subject to $z_{CC} \leq \frac{3}{2} z_{CD}$ and $z_{CC} \leq \frac{3}{2} z_{DC}$. Then

$$z_{CC} \leq \tfrac{3}{2} \min(z_{DC}, z_{CD}) \leq \tfrac{3}{2} \cdot \frac{1 - z_{CC}}{2} \quad \Longrightarrow \quad z_{CC} \leq \tfrac{3}{7},$$

with equality iff $z_{DC} = z_{CD} = \frac{2}{7}$. The optimal correlated equilibrium is

$$z^* = (z_{DD}, z_{DC}, z_{CD}, z_{CC}) = \left(0,\ \tfrac{2}{7},\ \tfrac{2}{7},\ \tfrac{3}{7}\right),$$

with total payoff $\frac{80}{7}$ and $\frac{40}{7} \approx 5.71$ for each drone. The controller cannot put more weight on (C, C): a drone told C must believe the other drone is likely enough to dare that yielding is optimal. Compared with yielding, daring gains 2 when the other yields (8 vs. 6) and loses 3 when the other dares (0 vs. 3), so obedience requires $2z_{CC} \le 3z_{CD}$ (and $2z_{CC} \le 3z_{DC}$ for the other drone); both constraints bind at z^*.

Rubric (6 pts).

- 1 pt: correct objective; 1 pt: argue $z_{DD} = 0$.
- 2 pts: optimal z^* (full credit also for a correct LP-solver output accompanied by a verification that all four constraints hold and two bind).
- 1 pt: payoffs 40/7 each. 1 pt: explanation of the binding constraint.

(d) Yes. A drone told D knows the other was told C; daring gives $8 > 6$. A drone told C assigns probability $\frac{1}{2}$ to each of D and C for the other drone: yielding gives $\frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 6 = 4.5$ and daring gives $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 8 = 4$, so it obeys. Each drone's payoff is $\frac{1}{3}(8 + 3 + 6) = \frac{17}{3} \approx 5.67$: higher than in the mixed equilibrium (4.8) and fairer than either pure equilibrium (8 vs. 3), but slightly below the optimum $\frac{40}{7} \approx 5.71$ from (c), because it puts less weight on (C, C) than the constraints allow.

Rubric (4 pts).

- 2 pts: both obedience checks. 1 pt: payoff 17/3. 1 pt: comparison.

Problem 2 (20 points)

(a)

t	losses ℓ_t	weights w_t	probabilities p_t	expected loss
1	$(1, 0, 1)$	$(1, 1, 1)$	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	$\frac{2}{3}$
2	$(0, 1, 1)$	$(\frac{1}{2}, 1, \frac{1}{2})$	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$	$\frac{3}{4}$
3	$(1, 1, 0)$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{4})$	$(\frac{2}{5}, \frac{2}{5}, \frac{1}{5})$	$\frac{4}{5}$
4	$(0, 1, 0)$	$(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	$\frac{1}{3}$

Total expected loss $= \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \frac{1}{3} = \frac{51}{20} = 2.55$. (Weights after round 4 would be $(\frac{1}{4}, \frac{1}{8}, \frac{1}{4})$.)

Rubric (8 pts).

- 3 pts: weights (1 pt per row for rows 2–4).
- 2 pts: probabilities.
- 2 pts: per-round expected losses. 1 pt: total 51/20.
- A common error is updating with $\beta^{1-\ell}$ (rewarding mistakes): at most 3 of 8.

(b) Cumulative losses of the forecasters are $(2, 3, 2)$, so the best forecaster has loss 2 and the regret is $51/20 - 2 = 11/20 = 0.55$. With $\eta = \ln 2$, $n = 3$, $T = 4$ the bound is $\ln 3 / \ln 2 + (4 \ln 2)/8 \approx 1.585 + 0.347 = 1.93$; the realized regret is well within the guarantee (the bound is a worst case over all loss sequences).

Rubric (4 pts).

- 1 pt: best forecaster total 2. 1 pt: regret 11/20. 2 pts: bound ≈ 1.93 and comparison.

(c) Setting the derivative of $\frac{\ln n}{\eta} + \frac{\eta T}{8}$ to zero gives

$$\eta^* = \sqrt{\frac{8 \ln n}{T}}, \qquad \frac{\ln n}{\eta^*} + \frac{\eta^* T}{8} = \sqrt{\frac{T \ln n}{2}}.$$

With $n = 16$, $T = 10{,}000$: $\ln 16 \approx 2.773$, so

$$\eta^* = \sqrt{8 \cdot 2.773 / 10^4} \approx 0.0471, \qquad \text{Regret} \le \sqrt{5000 \cdot 2.773} \approx 117.7,$$

i.e., average regret at most ≈ 0.0118 per round.

Rubric (4 pts).

- 1 pt: formula for η^*. 1 pt: numeric η^*. 1 pt: bound ≈ 117.7. 1 pt: average ≈ 0.012.

(d) Take losses $\ell_1 = (\frac{1}{2}, 0)$, then alternate $\ell_t = (0, 1)$ for even t and $\ell_t = (1, 0)$ for odd $t \ge 3$. After round 1 the cumulative losses are $(\frac{1}{2}, 0)$, so FTL follows forecaster 2 and suffers loss 1; then the cumulative losses are $(\frac{1}{2}, 1)$, FTL switches to forecaster 1 and again suffers loss 1; and so on. FTL loses 1 in every round after the first, for a total of $T - \frac{1}{2}$, while each forecaster's total is about $T/2$. So FTL's regret is about $T/2$, linear in T.

FTL is deterministic and jumps completely to the current leader, so an adversary who knows the rule can always put the loss on the forecaster it is about to follow. Hedge randomizes, and its probabilities change smoothly: when the cumulative losses are nearly tied, it follows each forecaster with probability close to $\frac{1}{2}$ and loses about $\frac{1}{2}$ per round, matching the best forecaster. (No deterministic algorithm can have sublinear regret against an adaptive adversary.)

Rubric (4 pts).

- 2 pts: a valid sequence with the FTL loss computed.
- 2 pts: explanation (determinism/instability vs. randomization/smooth weights).

Problem 3 (20 points)

(a) Row plays H iff Column's count of H is at least its count of T (Row wants to match the more likely action; ties go to H). Column plays T iff Row's count of H strictly exceeds its count of T, and H otherwise (Column wants to mismatch; at a tie it is indifferent and plays H).

Round	Row's counts of Column (H, T)	Column's counts of Row (H, T)	Row plays	Column plays
1	$(1, 0)$	$(0, 1)$	H	H
2	$(2, 0)$	$(1, 1)$	H	H
3	$(3, 0)$	$(2, 1)$	H	T
4	$(3, 1)$	$(3, 1)$	H	T
5	$(3, 2)$	$(4, 1)$	H	T
6	$(3, 3)$	$(5, 1)$	H	T
7	$(3, 4)$	$(6, 1)$	T	T
8	$(3, 5)$	$(6, 2)$	T	T

Pattern: play moves around the best-response cycle $(H,H) \to (H,T) \to (T,T) \to (T,H) \to (H,H) \to \cdots$. At each stage the player who is "losing" switches only after the opponent's count has shifted enough, so the runs get longer over time (here runs of length 2, 4, and then a run of (T,T) that continues past round 8).

Rubric (8 pts).

- 1 pt per row for rows 2–7 (6 pts); row 8 is covered by the pattern description.
- 2 pts: description of the cycle with lengthening runs.
- Tie-breaking errors in rounds 2 and 6 are the most common; deduct only once if subsequent rows are consistent with the error.

(b) Matching pennies is a two-player zero-sum game, so by Robinson's theorem (1951) the empirical frequencies of fictitious play converge to the set of equilibrium (maximin) strategies. The unique equilibrium is $(\frac{1}{2}, \frac{1}{2})$ for each player, so each player's frequency of H tends to $\frac{1}{2}$. The value of the game is 0, and the players' average payoffs converge to it.

Rubric (4 pts).

- 1 pt: frequencies $\to (\frac{1}{2}, \frac{1}{2})$. 2 pts: Robinson's theorem (zero-sum). 1 pt: value 0.

(c) Suppose (a^*, b^*) is played in every round $t \geq t_0$. Row's belief q_t about Column (the empirical distribution) then converges to the point mass on b^*, since the fraction of rounds in which b^* was played tends to 1. Suppose, for contradiction, that some a' satisfies $u_1(a', b^*) > u_1(a^*, b^*)$. Expected payoffs are continuous (linear) in beliefs, so for all large t we have $u_1(a', q_t) > u_1(a^*, q_t)$, i.e., a^* is not a best response to q_t. This contradicts the fact that Row plays a^* in round t. Hence a^* is a best response to b^*; symmetrically b^* is a best response to a^*, and (a^*, b^*) is a pure Nash equilibrium.

Rubric (4 pts).

- 1 pt: beliefs converge to the point mass. 2 pts: continuity/contradiction step. 1 pt: both players covered.

(d) No. Matching pennies has no pure equilibrium, so by (c) play cannot settle on a pure profile; it cycles forever with ever-longer runs. The empirical *frequencies* nevertheless converge to the mixed equilibrium $(\frac{1}{2}, \frac{1}{2})$. Convergence of frequencies is a statement about time averages; each round, each player still plays a pure best response, and nobody is ever actually randomizing $\frac{1}{2}$–$\frac{1}{2}$.

Rubric (4 pts).

- 2 pts: play does not converge (with a reason). 2 pts: clear distinction between day-to-day play and time averages.

Problem 4 (20 points)

(a) Rosenthal's potential sums each resource's cost function up to its load:

$$\Phi(n_A, n_B) = \sum_{k=1}^{n_A} k + \sum_{k=1}^{n_B} (k+1) = \frac{n_A(n_A+1)}{2} + \frac{n_B(n_B+3)}{2}.$$

If an analyst moves from A to B, her cost changes from $c_A(n_A) = n_A$ to $c_B(n_B+1) = n_B + 2$, and Φ changes by $-c_A(n_A) + c_B(n_B+1)$ (the last term of the A-sum is removed and a new term is added to the B-sum). These are equal. The move from B to A is symmetric.

Rubric (4 pts).

- 2 pts: correct Φ. 2 pts: verification for a switch (one direction suffices if the student notes symmetry).

(b) An analyst on A does not switch iff $n_A \le n_B + 2$; an analyst on B does not switch iff $n_B + 1 \le n_A + 1$, i.e., $n_B \le n_A$ (each condition applies only if the cluster is used). With $n_A + n_B = 4$:

(n_A, n_B)	A-user cost vs. switch	B-user cost vs. switch	Equilibrium?
$(0,4)$	—	5 vs. 1	no
$(1,3)$	1 vs. 5	4 vs. 2	no
$(2,2)$	2 vs. 4	3 vs. 3	yes
$(3,1)$	3 vs. 3	2 vs. 4	yes
$(4,0)$	4 vs. 2	—	no

The equilibria are the load vectors $(2,2)$ and $(3,1)$. Counting named analysts: $\binom{4}{2} + \binom{4}{3} = 6 + 4 = 10$ equilibrium profiles.

Rubric (6 pts).

- 2 pts: correct no-switch conditions. 2 pts: both load vectors (1 pt each). 2 pts: count of 10.

(c) From $(0,4)$ with $\Phi = 14$: a B-user's cost is 5, and switching to A costs 1, so one switches, giving $(1,3)$ with $\Phi = 10$. A B-user now pays 4 and would pay 2 on A, so one switches, giving $(2,2)$ with $\Phi = 8$. At $(2,2)$ no analyst can strictly improve (B-users: 3 vs. 3; A-users: 2 vs. 4), so the dynamics stop at the equilibrium $(2,2)$. Each step lowers Φ by exactly the mover's cost reduction (4, then 2).

Rubric (4 pts).

- 2 pts: correct path. 1 pt: Φ values $14, 10, 8$. 1 pt: stopping point.

(d) Social cost $SC = n_A \cdot n_A + n_B(n_B + 1)$: for $n_A = 0, 1, 2, 3, 4$ it is $20, 13, 10, 11, 16$. The optimum is $(2,2)$ with cost 10. The worst equilibrium $(3,1)$ costs 11 and the best equilibrium $(2,2)$ costs 10, so

$$\text{PoA} = \tfrac{11}{10}, \qquad \text{PoS} = 1.$$

Rubric (6 pts).

- 2 pts: social cost values. 1 pt: optimum $(2,2)$. 2 pts: PoA $= 11/10$. 1 pt: PoS $= 1$.

Problem 5 (20 points)

Each part: 1 pt for the correct letter, 3 pts for the justification.

5.1 True. Let G' be G plus a new edge. Then

$$C(\text{eq}_{G'}) \le \tfrac{4}{3}\,\text{OPT}(G') \le \tfrac{4}{3}\,\text{OPT}(G) \le \tfrac{4}{3}\,C(\text{eq}_G).$$

The first inequality is the $4/3$ price-of-anarchy bound; the second holds because the planner can ignore the new edge; the third because the equilibrium cannot beat the optimum. The Braess network with affine latencies shows that the factor $4/3$ is attained.

5.2 True. Let $\bar{x}, \bar{y}$ be the time averages and $\bar{u}$ Row's average realized payoff. Row's regret bound gives $\max_x u(x, \bar{y}) \le \bar{u} + \varepsilon_T$, and Column's gives $\bar{u} \le \min_y u(\bar{x}, y) + \varepsilon_T$. Hence

$$\max_x u(x, \bar{y}) - \min_y u(\bar{x}, y) \le 2\varepsilon_T, \qquad \text{while} \qquad \min_y u(\bar{x}, y) \le u(\bar{x}, \bar{y}) \le \max_x u(x, \bar{y}),$$

so neither player can gain more than $2\varepsilon_T$ by deviating from $(\bar{x}, \bar{y})$.

5.3 False. No-regret dynamics only guarantee convergence to the set of *coarse correlated* equilibria, which is generally much larger than the set of Nash equilibria. Counterexample: in the coordination game with payoff 1 to both players if they choose the same action in $\{A, B\}$ and 0 otherwise, let each player use "alternate $A, B, A, B, \ldots$ as long as my regret is ≤ 0, and switch to Hedge forever if it ever becomes positive." This algorithm has sublinear regret against every sequence. In self-play both players earn the maximum payoff in every round, so neither switches, and the empirical distribution is $\frac{1}{2}(A, A) + \frac{1}{2}(B, B)$. This is a correlated distribution, not the product of any mixed strategies, so it is not a Nash equilibrium; indeed it stays at a fixed positive distance from each of the three equilibria (A, A), (B, B) and $(\frac{1}{2}, \frac{1}{2}) \otimes (\frac{1}{2}, \frac{1}{2})$.

5.4 True. A CE requires, for each recommended action a_i and each deviation a_i', that

$$\textstyle\sum_{a_{-i}} z(a_i, a_{-i}) \left[u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) \right] \ge 0.$$

The CCE constraint for a fixed deviation a' is the sum of these CE constraints over a_i (with $a_i' = a'$ in each), so it holds whenever all CE constraints hold. (The converse fails: CCE only guards against deviations chosen before seeing the recommendation.)

5.5 False. Computing a Nash equilibrium of a two-player game is PPAD-complete (Chen and Deng, 2006), and no polynomial-time algorithm is known. The Lemke–Howson algorithm always finds an equilibrium but can take exponentially many steps. (Zero-sum games, in contrast, can be solved in polynomial time by linear programming.)

Rubric (20 pts).

- Per part: 1 pt letter; 3 pts justification. For 5.1, full justification needs the chain of three inequalities. For 5.2, the two regret inequalities earn 2 pts, and combining them earns 1 pt. For 5.3, naming CCE earns 2 pts and a concrete example or a precise argument earns 1 more. For 5.4, the summation argument is required for 3 pts. For 5.5, "PPAD-complete" earns 2 pts, and the zero-sum contrast or the Lemke–Howson remark earns 1 more.